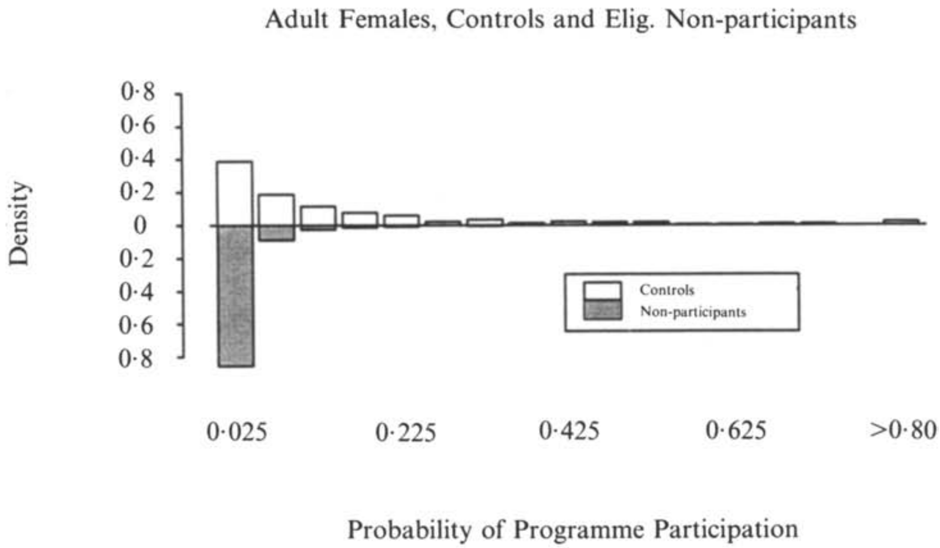


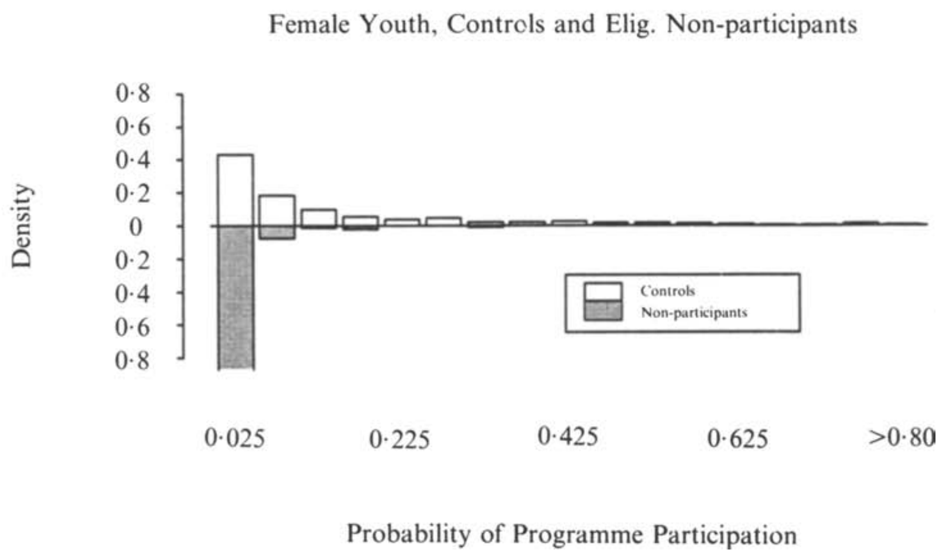


FIGURE 1. —continued.

demonstrate that different supports and differences in the density of $P$ in comparison and control samples are major sources of evaluation bias as conventionally measured.

## 9. DECOMPOSING THE CONVENTIONAL MEASURE OF EVALUATION BIAS

It is instructive to decompose the conventional measure of evaluation bias into components due to selection bias, rigorously defined, components due to mismatching or misweighting of the data and an additional source of discrepancy between experimental and nonexperimental estimates of programme impact that arises from the differences in supports over which experimental and nonexperimental treatment effects are estimated. Let $B$ be the conventional measure of bias arising from using mean comparison group ($D=0$) outcomes to proxy mean participant ($D=1$) outcomes. It is analogous to (8) but does not condition on $X$: $B=E(Y_0|D=1)-E(Y_0|D=0)$. To estimate this measure of bias, we use the randomized-out controls from the JTPA experimental data to construct $E(Y_0|D=1)$ combined with data from a nonexperimental comparison group to construct $E(Y_0|D=0)$. We investigate a variety of comparison groups, starting with the ENPs.

$S_1$ is the support of $X$ for $D=1$, $S_0$ is the support of $X$ for $D=0$, and $S_{10}$ is the region in which the supports overlap. Denote the region contained in $S_0$ but not in $S_{10}$ by $S_0\backslash S_{10}$ and the region contained in $S_1$ but not in $S_{10}$ by $S_1\backslash S_{10}$. Bias $B$ can be broken down into components due to differing supports of $X$, to differing distributions of $X$ over the same support in the two populations and to differences in outcomes that are present even after controlling for observables. Assuming for simplicity that the $X$ are continuously distributed,

$$\begin{aligned}B&=\int_{S_1} E(Y_0|X, D=1)f(X|D=1)dX-\int_{S_0} E(Y_0|X\ D=0)f(X|D=0)dX\\&=B_1+B_2+B_3,\end{aligned}$$

where

$$B_1=\left\{\int_{S_1\backslash S_{10}} E(Y_0|X, D=1)f(X|D=1)dX-\int_{S_0\backslash S_{10}} E(Y_0|X, D=0)f(X|D=0)dX\right\},$$

$$B_2=\int_{S_{10}} E(Y_0|X, D=0)\{f(X|D=1)-f(X|D=0)\}dX,$$

$$B_3=\int_{S_{10}} \{E(Y_0|X, D=1)-E(Y_0|X, D=0)\}f(X|D=1)dX.$$

The first component of bias ($B_1$) arises because of nonoverlapping support. For some participants there are no comparable nonparticipants and for some nonparticipants there are no comparable participants. The second component ($B_2$) arises from different distributions of $X$ within the two populations. The third component ($B_3$) is due to differences in outcomes that remain even after conditioning on observables and making comparisons on a region of common support. This component is due to selection on unobservables and is selection bias as rigorously defined in econometrics. (See Heckman, Ichimura, Smith and Todd (1996a, b).) The selection bias that results after conditioning on $X$ and weighting the data comparably is

$$\bar{B}_{S_X}=\frac{\int_{S_{10}}\{E(Y_0|X,D=1)-E(Y_0|X,D=0)\}f(X|D=1)dX}{\int_{S_{10}}f(X|D=1)dX}.$$

Weighting by $f(X|D=1)$ is appropriate since we seek to investigate the bias for $M(S)$ which is defined for the group $D=1$. Observe that $B_3=(\bar{B}_{S_X})\int_{S_{10}}f(X|D=1)dX$.

Exploiting our access to data on randomized-out controls allows us to estimate the bias $B$ by the difference between control and comparison group mean outcomes

$$\hat{B}=\frac{1}{N_1^*}\sum_{i\in I_1^*}Y_{0i}^1-\frac{1}{N_0}\sum_{j\in I_0}Y_{0j}^0,$$

where $Y_{0i}^1$ is the outcome of randomized-out applicants ($D=1$, $R=0$) and $Y_{0j}^0$ is the outcome of non-applicants in the comparison group ($D=0$). $N_1^*$ and $N_0$ are the sample sizes in the two groups. $I_1$ is the set of indices for eligible and provisionally-accepted applicants ($D=1$), $I_1^*$ is a subset of $I_1$ corresponding to randomized-out accepted applicants ($D=1, R=0$), $I_0$ is a set of indices for members of the comparison group ($D=0$). Estimating the relative contribution of each of these components to the total bias as conventionally measured, we find that the first and second components contribute the most. The selection bias component $B_3$ is only a small component of evaluation bias as conventionally measured.

To assess the empirical importance of each of the potential sources of bias, we perform the following decompositions. The components of $B$ are defined conditional on $P$ rather than $X$. First condition on $P(X)$ to obtain

$$E(Y_0|X,D=1)=E(Y_0|P(X),D=1)+V_1,$$
$$E(Y_0|X,D=0)=E(Y_0|P(X),D=0)+V_0,$$

where $V_1$ and $V_0$ are orthogonal to $P(X)$. Denote the values of $Y_0$ for $D=1$ with associated value $P$ by $Y_0^1(P)$ and the value of $Y_0$ for $D=0$ by $Y_0^0(P)$. Let $E(Y_0|P,D=0)$ be the local linear regression estimator of $\hat{E}(Y_0|P,D=0)$. (See Fan (1992) or Heckman, Ichimura, Smith and Todd (1996b, c) for detailed discussion of local linear regression methods.) Then the estimator for $B$ is

$$\hat{B}=\frac{1}{N_1^*}\sum_{i\in I_1^*}Y_0^1(P_i)-\frac{1}{N_0}\sum_{j\in I_0}Y_0^0(P_j).$$

The orthogonal components $V_1$ and $V_0$ are irrelevant for computing the bias since they average out to zero. Then we may decompose $\hat{B}$ in the following way, where we use the best-predicting $P(X)$ model for each demographic group to define the support

$$\begin{aligned}\hat{B}=&\left[\frac{1}{N_1^*}\sum_{i\in I_1^*(S_1\backslash S_{10})}Y_0^1(P_i)-\frac{1}{N_0}\sum_{j\in I_0(S_0\backslash S_{10})}Y_0^0(P_j)\right]\\&+\left[\frac{1}{N_1^*}\sum_{i\in I_1^*(S_{10})}[\hat{E}(Y_0|P_i,D=0)]-\frac{1}{N_0}\sum_{j\in I_0(S_{10})}Y_0^0(P_j)\right]\\&+\left[\frac{1}{N_1^*}\sum_{i\in I_1^*(S_{10})}[Y_0^1(P_i)-\hat{E}(Y_0|P_i,D=0)]\right],\end{aligned}$$

TABLE 2

*Decomposition of difference in post-programme mean earnings*
*Bootstrapped standard errors shown in parentheses‡*
*Percentage of mean difference attributable to components in square brackets*
*Earnings measured in average monthly dollars*

| Experimental Controls and eligible nonparticipants (ENPs)† | | | | | | |
|---|---|---|---|---|---|---|
| | Mean difference $\hat{B}$ | Non-overlap* $\hat{B}_1$ | Density weighting $\hat{B}_2$ | Selection bias $\hat{B}_3$ | Average bias $(\hat{\hat{B}}_{S_P})$ | Selection bias** $(\hat{\hat{B}}_{S_P})$ as a % of treatment impact |
| Adult males | −342 | 218 | −584 | 23 | 38 | 87% |
| (std. err.) [%] | (47) | (38) [−64%] | (41) [170%] | (33) [−7%] | (63) | |
| Adult females | 33 | 80 | −78 | 31 | 38 | 129% |
| (std. err.) [%] | (26) | (13) [242%] | (18) [−235%] | (26) [94%] | (33) | |
| Male youth | 20 | 142 | −131 | 9 | 14 | 23% |
| (std. err.) [%] | (57) | (28) [704%] | (35) [−650%] | (42) [46%] | (64) | |
| Female youth | 42 | 74 | −67 | 35 | 49 | 7239% |
| (std. err.) [%] | (36) | (17) [177%] | (26) [−161%] | (28) [84%] | (42) | |

| Experimental Controls and SIPP Eligibles†† | | | | | | |
|---|---|---|---|---|---|---|
| | Mean difference $\hat{B}$ | Non-overlap* $\hat{B}_1$ | Density weighting $\hat{B}_2$ | Selection bias $\hat{B}_3$ | Average bias $(\hat{\hat{B}}_{S_P})$ | Selection bias** $(\hat{\hat{B}}_{S_P})$ as a % of treatment impact |
| Adult males | −145 | 151 | −417 | 121 | 192 | 440% |
| (std. err.) [%] | (56) | (30) [−104%] | (44) [287%] | (33) [−83%] | (57) | |
| Adult females | 47 | 97 | −172 | 122 | 198 | 676% |
| (std. err.) [%] | (23) | (19) [206%] | (16) [−367%] | (15) [260%] | (26) | |
| Male youth | −188 | 65 | −263 | 9 | 21 | 36% |
| (std. err.) [%] | (106) | (108) [−35%] | (53) (139%] | (25) [−5%] | (90) | |
| Female youth | −88 | 83 | −168 | −3 | −13 | 1969% |
| (std. err.) [%] | (38) | (22) [−94%] | (27) [191%] | (10) [3%] | (58) | |

‡ They are based on 50 replications of the data with 100% sampling.

† The best predictor Control-ENP participation models for all the demographic groups include indicator variables for site, age, race, education, marital status, children less than 6 and labour force transitions. In addition to these variables, the adult male model also includes an indicator for vocational training history, the number of household members, earnings in the month of random assignment or eligibility determination (RA or EL) and number of jobs held in 18 months before RA or EL. The adult female model includes an indicator for recent schooling, earnings in the month of RA or EL and number of labour force transitions in the 24 months prior to RA or EL. The male youth model includes average earnings in the 6 months and 12 months prior to RA or EL and average positive earnings in the 6 months before RA or EL. The female youth model includes earnings in the 12 months before RA or EL.

†† The best predictor Control-SIPP model includes indicators for age, race, education, marital status, children age less than 6, labour force transition patterns and levels of earnings in the preceding year. The data used are SIPP eligibles and experimental JTPA controls.

* A 2% trimming rule was used for adult males and females and a 5% trimming rule for youth was used in determining the overlapping support region (see Appendix C for a description of how the support is determined). The proportion of Controls and ENPs falling in the overlap region ($S_P$) are: 60% and 96% of adult males, 82% and 96% of adult females, 67% and 92% of male youth and 71% and 93% of female youth. The proportion of SIPPs and Controls falling in the overlap region are: 63% and 96% of adult males, 61% and 96% of adult females, 41% and 90% of male youth and 20% and 89% of female youth. A 0·06 fixed bandwidth and a biweight kernel, defined in Appendix A, were used for the nonparametric estimates.

** The final column displays the ratio of the absolute value of $(\hat{\hat{B}}_{S_P})$ to the absolute value of experimental impact estimate.